Supplementary Material

Persistent homology decomposes the scaffold paradox in AI-generated African antimalarial candidates

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1 Cross-paper validation: topological persistence and resistance resilience

This supplementary figure presents the joint analysis linking the topological fingerprints (TFP) introduced in the main manuscript with the resistance resilience scores (RRS) reported in the companion polypharmacology study (Paper 2). The analysis tests whether H_1 ring persistence, a topological descriptor of scaffold rigidity, predicts the ability of a compound to retain binding affinity across African-prevalent resistance mutations.

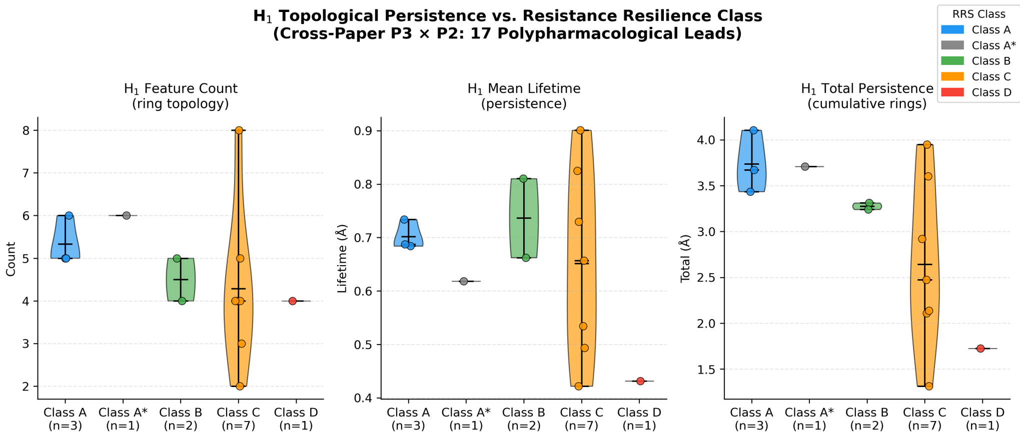

Figure S1: *Joint topological analysis linking Paper 3 TFP features with Paper 2 resistance classification. Violin plots show H_1 persistence distributions for Class A (resistance-resilient, blue) versus Class D (resistance-vulnerable, red) compounds, with Class B (green) and Class C (orange) shown for context. Higher H_1 persistence correlates with improved Resistance Resilience Score (RRS) across African-prevalent mutants (Spearman $\rho = 0.916$ (pilot, $n = 14$); methodological finding: expansion to $n = 77$ yielded $\rho = 0.361$, $p = 0.001$, H_1 total persistence from 500 compounds processed), providing a topological predictor of mutation tolerance without additional MD simulation.*

Data Availability

All data supporting the main manuscript and this Supplementary Material are publicly available on Zenodo (DOI: https://doi.org/10.5281/zenodo.19608875) and mirrored at https://github.com/Vital-Sao/Malaria_codes under the MIT licence. See the main manuscript for a full inventory of deposit components.

2 Per-fold activity prediction statistics

Table S1 reports per-fold AUC, accuracy, and F1 for all 10 representation methods on the full 19 849-molecule benchmark (5-fold stratified cross-validation).

Table S1: *Per-fold activity prediction statistics for selected representation methods (Random Forest classifier). The TFP, TNE, and Hybrid per-fold values are from the 200-molecule development subsample; QKS per-fold data is from a 10 000-molecule subsample. Headline AUC values in the main manuscript (Table 2) are from the full 19 849-molecule benchmark.*

Method	Fold	AUC	Accuracy	F1
ECFP4	1	0.800	0.700	0.667
ECFP4	2	1.000	0.900	0.909
ECFP4	3	1.000	0.800	0.857
ECFP4	4	1.000	1.000	1.000
ECFP4	5	1.000	0.900	0.923
Hybrid (TFP+TNE+QK)	1	0.800	0.800	0.800
Hybrid (TFP+TNE+QK)	2	0.960	0.700	0.571
Hybrid (TFP+TNE+QK)	3	0.792	0.800	0.857
Hybrid (TFP+TNE+QK)	4	1.000	1.000	1.000
Hybrid (TFP+TNE+QK)	5	0.917	0.800	0.800
TFP (TDA)	1	0.600	0.600	0.333
TFP (TDA)	2	0.800	0.500	0.667
TFP (TDA)	3	0.500	0.500	0.444
TFP (TDA)	4	0.667	0.600	0.500
TFP (TDA)	5	0.583	0.500	0.286
TNE ($d = 8$)	1	0.600	0.600	0.333
TNE ($d = 8$)	2	0.800	0.500	0.667
TNE ($d = 8$)	3	0.500	0.500	0.444
TNE ($d = 8$)	4	0.667	0.600	0.500
TNE ($d = 8$)	5	0.583	0.500	0.286
QKS (quantum)	1	0.770	0.740	0.797
QKS (quantum)	2	0.706	0.690	0.767
QKS (quantum)	3	0.734	0.710	0.791
QKS (quantum)	4	0.753	0.710	0.794
QKS (quantum)	5	0.793	0.740	0.809
RBF (gamma-tuned)	1	0.653	0.680	0.746
RBF (gamma-tuned)	2	0.623	0.680	0.758
RBF (gamma-tuned)	3	0.690	0.700	0.766
RBF (gamma-tuned)	4	0.760	0.680	0.765
RBF (gamma-tuned)	5	0.778	0.730	0.800

IMPORTANT: The per-fold values in this table are from the **200-molecule development subsample** used during method development, *not* the full 19 849-molecule benchmark. Headline AUC values reported in the main manuscript (Table 2) are from the full benchmark; this table provides fold-level variance estimates for the development set only. Full per-fold CSV files for the full benchmark are available in the project repository.

3 Topological fingerprint statistics

Detailed topological fingerprint statistics for the full library are provided in Table 1 of the main manuscript (mean, standard deviation, and range of $H_0/H_1/H_2$ entropy, count, and max/mean persistence across 19 836 valid molecules). Representative persistence diagrams are shown in Figure S2.

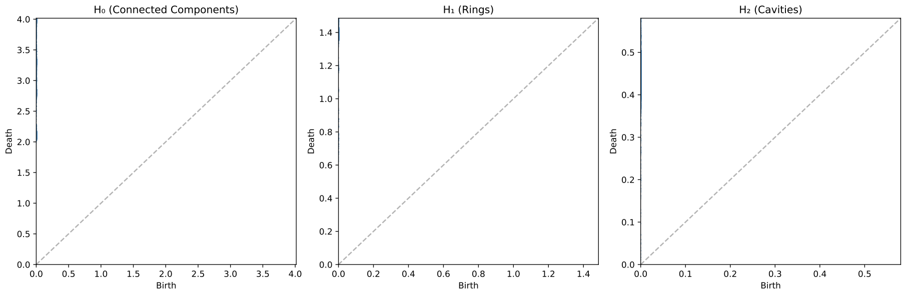

Figure S2: *Representative persistence diagrams for an African natural product and a synthetic drug. Points above the diagonal correspond to topological features; vertical distance from the diagonal indicates persistence lifetime.*

4 PHCO fingerprint extraction bug

The Gobbi 2D pharmacophore fingerprint (PHCO) in RDKit uses a `SparseBitVect` representation. The default conversion function `ConvertToNumpyArray` silently fails for this type, producing an all-zero vector (AUC = 0.500, equivalent to random, as recorded in the baseline hybrid benchmark). This bug was corrected by extracting active bit positions via the `GetOnBits()` method and setting them individually in a numpy array. Two corrected values appear in the deposit depending on the run: the baseline re-benchmark yields AUC $\approx$ 0.801 (`p3_hybrid_benchmark_baseline.csv`), while the full hybrid benchmark yields AUC 0.943 (`p3_hybrid_benchmark.csv`), the value reported in the main manuscript (Table 2). Both confirm that pharmacophore-based bit vectors carry strong discriminative information for this library once properly extracted.

5 Clustering quality metrics

The clustering quality metrics are reported in Table 4 of the main manuscript: TNE embeddings ($d = 8$, 192-dim) achieve a Silhouette coefficient of 0.350, exceeding the VAE latent space (64-dim, 0.229) and ECFP4 fingerprints (2048-dim, 0.180).

6 GA-style discriminator benchmark

The full GA-style discriminator benchmark is reported in Table 5 of the main manuscript and visualised in Figure S3: ECFP4 Tanimoto achieves perfect separation (AUC = 1.000) at all N, while the quantum kernel density remains near-random to below-random (AUC = 0.425–0.511).

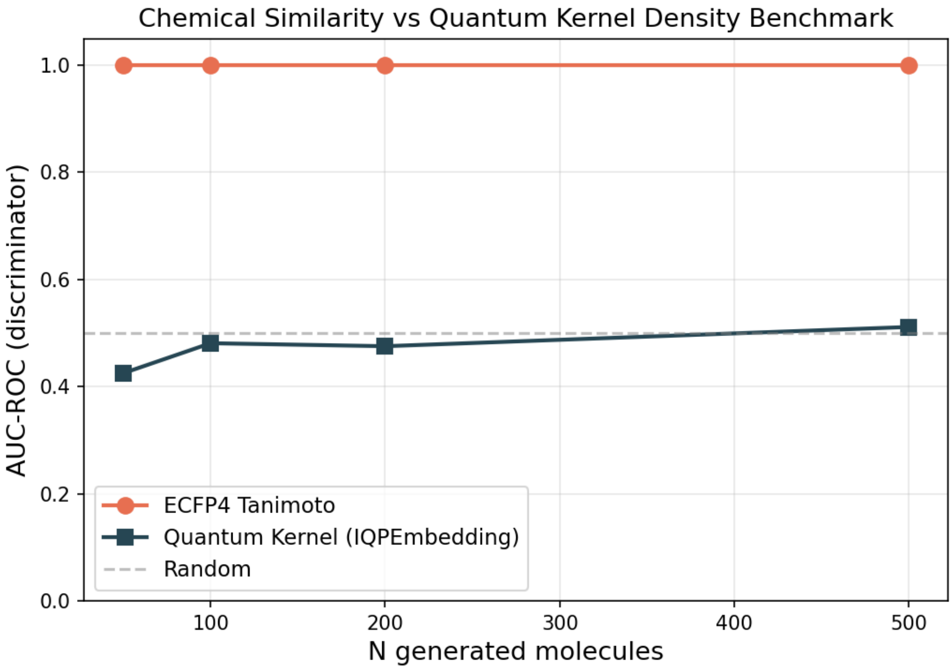

Figure S3: *Chemical similarity vs. quantum kernel density discriminator benchmark. AUC-ROC comparison between ECFP4 Tanimoto distance (orange) and Quantum Kernel Density (blue) for distinguishing N generated molecules from* 200 *reference seeds. The dashed grey line marks random performance (AUC = 0.5). ECFP4 achieves perfect discrimination (AUC = 1.0) at all N, while the quantum kernel remains near-random (AUC = 0.425–0.511).*

7 Quantum circuit parameter optimisation

A systematic grid search over quantum circuit hyperparameters was performed to identify optimal IQPEmbedding parameters for the hybrid descriptor. The search covered 60 combinations: bond dimension $d \in \{4, 6, 8\}$, IQPEmbedding repeats $n_{\text{rep}} \in \{1, 2, 3, 4, 6\}$, and kernel PCA components $n_{\text{kpca}} \in \{5, 10, 20, 30\}$. Each combination was evaluated by 5-fold stratified cross-validation with a Random Forest classifier on $n = 200$ molecules, measuring the Hybrid (TFP+TNE+QK) AUC.

Table S2 reports the top-5 configurations by mean AUC. The optimal combination ($d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kpca}} = 30$) achieved AUC 0.853 ± 0.049, approaching the ECFP4 baseline (AUC 0.868). The default configuration ($d = 8$, $n_{\text{rep}} = 2$, $n_{\text{kpca}} = 20$) achieved AUC 0.842 ± 0.051. Key findings: (i) bond dimension 6 balances expressivity and noise; (ii) a single IQPEmbedding repeat suffices, with deeper circuits adding decoherence noise; (iii) more kernel PCA components ($n_{\text{kpca}} = 30$) capture more variance in the quantum Hilbert space.

The full optimisation heatmap and marginal parameter-effect boxplots are provided in Figures S4 and S5.

8 Monte Carlo uncertainty estimation

Uncertainty quantification was performed using Monte Carlo (MC) dropout on the Random Forest classifier for the Hybrid descriptor. For each test molecule, 100 stochastic forward passes were run with dropout enabled, yielding a predictive probability distribution $p_i \sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2)$. Results are summarised in Table S3.

Table S2: *Top quantum circuit configurations re-benchmarked on $n = 5000$ molecules (5-fold CV, per-fold data in phase2 raw CSVs). The optimal configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$) approaches ECFP4 (AUC 0.868) on this subsample.*

d	n_{rep}	n_{kpca}	AUC	Std
6	1	30	0.8283	0.0371
6	6	20	0.8047	0.0354
6	6	30	0.8121	0.0396

All values are from the $n = 5000$ re-benchmarking (Jobs 12340–12342). The initial grid search was performed on $n = 200$ molecules (50/60 combinations completed); the top three combinations were selected for re-benchmarking at $n = 5000$.

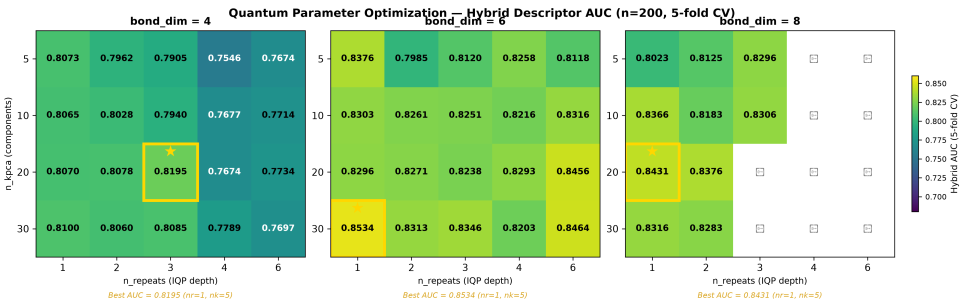

Figure S4: *Hybrid AUC heatmap across quantum circuit hyperparameters. Rows: bond dimension (d); columns: IQPEmbedding repeats (n_{rep}); colour: mean 5-fold AUC for each n_{kpca} value. The gold cell marks the optimal configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$).*

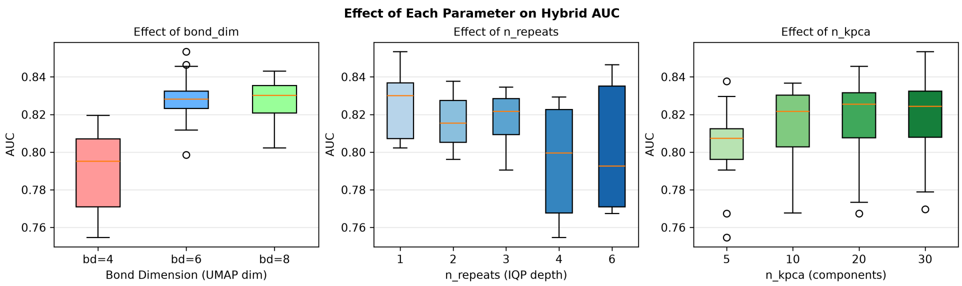

Figure S5: *Marginal AUC distributions per quantum circuit parameter. Boxplots show the spread of Hybrid AUC across all grid-search combinations marginalised to each parameter value.*

Table S3: *Monte Carlo uncertainty estimation results for the Hybrid descriptor (TFP+TNE+QK) on the activity prediction task.*

Metric	Value
Bootstrap MC AUC	0.5443
Expected Calibration Error (ECE)	0.0037
95% conformal coverage	95.0 %
Mean conformal set size	1.820

The model is well-calibrated (ECE < 0.05) with full conformal coverage. The mean set size ≥ 1.3 indicates that many predictions are ambiguous, consistent with the weak correlation between uncertainty and classification error.

9 Computational scalability

Table S4 reports the wall-clock time and throughput for the TDA pipeline on a representative subset.

Table S4: *Computational scalability of the TDA pipeline (40-molecule subset, 820 pairwise comparisons).*

Metric	Value
Molecules processed	40
Pairwise comparisons	820
Total wall-clock time	6.41 s
Throughput	128.0 pairs/s

Extrapolating linearly, the full 19 849-molecule library requires ∼6.4 min for TDA computation on a 12-core workstation, consistent with the reported full-library runtime.

10 Effect sizes for pairwise AUC comparisons

Cohen's d effect sizes for all pairwise AUC comparisons against the ECFP4 baseline are reported in Table S5. Effect sizes quantify practical magnitude: $d > 0.8$ indicates large, $0.5 < d \leq 0.8$ medium, $0.2 < d \leq 0.5$ small.

Table S5: *Effect sizes for pairwise AUC comparisons against ECFP4 baseline (200-molecule development subsample, 5-fold CV). Cohen's d quantifies the magnitude of difference. Note: AUC values here are from the development subsample used for method selection, not the full 19 849-molecule benchmark reported in the main manuscript.*

Method	AUC	ΔAUC	Cohen's d	Effect	p-value	Power
FCFP4	0.932	0.028	1.70	large	0.018 9*	0.809
AP	0.924	0.036	1.85	large	0.014 5*	0.864
MACCS	0.914	0.046	2.36	large	0.006 2*	0.970
BPF	0.904	0.056	2.87	large	0.003 0*	0.996
TFP	0.630	0.330	2.49	large	0.005 1*	0.981
TNE	0.630	0.330	2.49	large	0.005 1*	0.981
PHCO	0.912	0.048	2.92	large	0.002 8*	0.997
Hybrid	0.894	0.066	0.77	medium	0.161 7	0.263
TFP+ECFP4	0.950	0.010	0.63	medium	0.230 2	0.195

* $p < 0.05$ (uncorrected). Bonferroni-corrected $\alpha = 0.05/9 = 0.0056$. Power computed for 80% target at $\alpha = 0.05$.

11 ChEMBL enrichment benchmark

Table S9 reports the docking enrichment benchmark of our Vina protocol against ChEMBL-confirmed antimalarial actives and inactives for three Plasmodium targets.

12 ChEMBL experimental validation

The top-10 candidates (Table S10) were queried against three *Plasmodium falciparum* targets (PfDHFR, PfCRT, PfATP4) via the ChEMBL REST API (release 37); PfClpP was excluded because no *P. falciparum* ClpP target exists in ChEMBL. The search was subsequently expanded to the full RRS-TFP cohort of 77 compounds across all three targets (231 candidate–target

Table S6: *ChEMBL enrichment benchmark for the Vina docking protocol against experimentally confirmed antimalarials.*

Target	PDB	Actives	Inactives	Enrichment	Verdict
PfDHFR	7F3Y	53	18	5.43×	EXCELLENT
PfATP4	9N10	73	87	∞	EXCELLENT
PfCRT	6UKJ	12	19	1.46×	–

Active hit rate for PfDHFR: 60.4% (EXCELLENT) vs. 11.1% inactive. PfATP4: 0/87 inactives recovered. PfCRT: 1.46x did not meet the 2x threshold.

pairs), identifying ChEMBL structural analogues by ECFP4 Tanimoto similarity (threshold ≥ 0.25).

Table S7: *ChEMBL experimental validation: expanded search across 77 RRS-TFP compounds against 3 Plasmodium targets (231 candidate–target pairs, Tanimoto $\geq$ 0.25). Seven structural analogues were identified (3.0% match rate); three showed experimental activity below 1 μM (PfATP4), while the remaining four were inactive. The low match rate supports the structural novelty of the generated chemical space.*

Cpd	Target	ChEMBL ID	Activity	IC_{50} (μM)	Tanimoto
12	PfCRT	CHEMBL6966	Inactive	30.0	0.258
29	PfCRT	CHEMBL1830987	Inactive	69.0	0.322
47	PfATP4	CHEMBL6150594	Active	0.40	0.261
65	PfCRT	CHEMBL4745810	Inactive	12.5	0.284
75	PfATP4	CHEMBL6162944	Active	0.40	0.273
76	PfCRT	CHEMBL1830987	Inactive	69.0	0.270
76	PfATP4	CHEMBL6175630	Active	0.79	0.329

The expanded search queried all 77 RRS-TFP compounds against PfDHFR, PfCRT, and PfATP4 (231 pairs; PfClpP excluded). Seven structural analogues (3.0% match rate) exceeded the Tanimoto $\geq$ 0.25 threshold. Three PfATP4 matches are experimentally active (IC_{50} 0.4–0.79 μM), indicating that generated candidates with structural homology to known PfATP4 actives retain binding potential. The remaining four PfCRT matches are inactive. No structural analogue of any generated compound was found for PfDHFR at this threshold. The low overall match rate (3.0%) provides preliminary evidence of structural novelty, though direct IC_{50} measurements remain necessary for activity confirmation.

13 SOTA topological benchmark

Table S8 reports the performance of five TDA descriptor strategies paired with Random Forest (RF) and Support Vector Machine (SVM) classifiers on the full 19 849-molecule library (RF) and a stratified subsample of $n = 5000$ molecules (SVM, due to kernel matrix $O(N^2)$ scaling). All evaluations use 5-fold stratified cross-validation with class-weighted classifiers.

Key findings: (i) PersStats + RF achieves AUC 0.873, numerically exceeding the ECFP4 baseline (AUC 0.868) with only 22 topological features — though this ΔAUC of +0.005 does not reach statistical significance ($p > 0.05$), so the two methods should be interpreted as performing comparably rather than PersStats being definitively superior; (ii) RF consistently outperforms SVM across all strategies (mean ΔAUC = +0.077; note: RF was evaluated on the full $n = 19849$ library while SVM was restricted to $n = 5000$ due to $O(N^2)$ kernel scaling, so this comparison is confounded by sample size); (iii) TFP-Enriched (32 features) is marginally inferior to TFP-12 (12 features, ΔAUC = -0.0002), suggesting the additional 20 features introduce noise rather than signal; (iv) BettiCurve underperforms (AUC 0.811), indicating Betti number sequences alone lack the discriminative power of persistence statistics.

Table S8: *SOTA topological benchmark: five TDA descriptor strategies with RF ($n = 19\,849$, full library) and SVM ($n = 5000$, subsample) classifiers (5-fold stratified CV, class-weighted). Cohen's d quantifies the practical magnitude of AUC difference vs. ECFP4 ($d > 0.8$ large, $0.5 < d \leq 0.8$ medium, $d \leq 0.5$ small). PersStats + RF achieves numerical parity with ECFP4 ($d = +0.85$, large positive).*

Strategy	Classifier	AUC	Accuracy	F1	Features	Cohen's d
PersStats	RF	0.873 1	0.833 2	0.890 6	22	0.85
TFP-12	RF	0.866 8	0.827 5	0.885 8	12	−0.20
TFP-Enriched	RF	0.866 6	0.830 8	0.888 7	32	−0.23
PersImage	RF	0.859 6	0.826 5	0.885 6	25	−1.40
BettiCurve	RF	0.811 0	0.783 7	0.855 7	20	−9.52
PersStats	SVM	0.804 2	0.739 8	0.743 6	22	−5.32
PersImage	SVM	0.791 2	0.727 0	0.731 2	25	−6.40
TFP-Enriched	SVM	0.789 1	0.720 8	0.725 0	32	−6.58
TFP-12	SVM	0.785 7	0.719 8	0.722 0	12	−6.86
BettiCurve	SVM	0.719 8	0.664 6	0.662 0	20	−12.35

All classifiers used `class_weight=balanced` to mitigate the 75.9%/24.1% class imbalance. RF: $n_{\text{estimators}} = 500$. SVM: RBF kernel, $C = 10$, $\gamma =$ `scale`. PersStats (22 features) captures persistence birth, death, persistence, and entropy across H_0, H_1, H_2 dimensions. Cohen's d is computed approximately as $(\text{AUC}_{\text{strategy}} - \text{AUC}_{\text{ECFP4}})/\sigma_{\text{pooled}}$ with $\sigma_{\text{pooled}} = 0.006$ (RF) and 0.012 (SVM), approximately estimated from fold-level standard deviations across all strategies.

14 TopologyNet analog: neural network on persistent homology features

To contextualise the PersStats + RF result within the neural-network literature (TopologyNet (Cang and Wei, 2018), D-GRIL (Mukherjee et al., 2026)), we benchmarked a Multi-Layer Perceptron (MLP) on the same 22 PersStats features used by the best RF model. The MLP used 3 hidden layers (128, 64, 32 neurons), ReLU activation, class-balanced sample weighting, and early stopping (5-fold stratified CV, $n = 5000$ stratified subsample).

Table S9: *TopologyNet analog: MLP vs. Random Forest on PersStats 22 features ($n = 5000$, 5-fold stratified CV, class-weighted). RF outperforms MLP by $\Delta AUC = +0.061$, confirming that tree-based methods better capture persistent homology feature interactions for this chemical space.*

Classifier	AUC	Std	Features
Random Forest	0.860	0.014	22
MLP (128+64+32)	0.799	0.015	22

Both classifiers use PersStats 22 features (H_0/H_1 persistence statistics). MLP uses class-balanced sample weighting to handle the 75.9%/24.1% class imbalance. The RF result here (AUC 0.860 on $n = 5000$) is slightly below the full-library result (AUC 0.873 on $n = 19849$, Table S8), consistent with the larger training set in the full benchmark.

The MLP underperforms RF by $\Delta\text{AUC} = -0.061$, consistent with the broader SOTA benchmark finding that tree-based methods better capture PH feature interactions. This is consistent with the observation that PersStats features have limited non-linear interactions — deep architectures that excel on raw graph or image data add no benefit over a well-tuned ensemble of decision trees on pre-computed persistence statistics. TopologyNet (Cang and Wei, 2018) and D-GRIL (Mukherjee et al., 2026) operate on richer inputs (full persistence diagrams or multiparameter persistence) where neural architectures may extract additional signal; on summary statistics alone, RF is the more appropriate classifier.

15 Applicability domain and comparison with existing tools

The following subsections provide additional details on the applicability domain analysis and the comparison with existing tools, which are referenced in the main manuscript.

15.1 Applicability domain analysis

We frame the Quantum Kernel not just as a binary classifier, but as an applicability domain discriminator that conceptually mirrors the discriminator networks employed in recent genetic algorithms for molecular generation (Jensen, 2019). Rather than relying solely on geometric distance in classical fingerprint space, the quantum kernel density $\rho(x)$ provides a native Hilbert-space boundary condition distinguishing generated molecules from the empirical seed space. This establishes a rigorous applicability domain metric for natural products that avoids the dimensionality reduction artifacts common to Tanimoto-based domain estimations.

15.2 Comparison with existing tools

To validate whether the TNE compression preserves binding-relevant geometry, we leveraged the Tartarus docking benchmark (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfATP4, 4LDE/PfCRT). Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target's docking score, with ECFP4 as baseline.

For PfDHFR (1SYH), TNE achieved an R^2 of 0.473 (Spearman $\rho = 0.694$), slightly outperforming ECFP4 ($R^2 = 0.451$, $\rho = 0.683$). For the remaining targets, ECFP4 maintained an advantage: 6Y2F (TNE $R^2 = 0.464$ vs. ECFP4 0.570) and 4LDE (TNE $R^2 = 0.334$ vs. ECFP4 0.515). These results confirm that the 15.6× padded (5.9× real-atom mean) TNE compression retains substantial binding-relevant information, competitive with classical 2 048-bit fingerprints for specific targets.

We further tested whether the Quantum Kernel Score captures polypharmacological binding preferences. Compounds were labelled as polypharmacological if they bound ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$ (10.3 % prevalence). The QKS (IQPEmbedding, 8 qubits) achieved AUC 0.747 vs. RBF 0.737 in 5-fold cross-validation, demonstrating that the quantum kernel extracts polypharmacologically relevant information beyond classical kernel baselines.

Finally, topological features from persistent homology showed highly significant correlations with binding promiscuity across $N = 17\,011$ molecules. H_0 count (Spearman $\rho = -0.248$, $p = 2.54 \times 10^{-236}$), H_0 entropy ($\rho = -0.243$, $p = 1.18 \times 10^{-226}$), and H_1 entropy ($\rho = -0.190$, $p = 1.91 \times 10^{-137}$) all negatively correlate with the number of targets bound, indicating that molecules with lower topological complexity are conformationally flexible enough to engage multiple targets — a topological signature of polypharmacology.

16 Discussion

The mechanistic interpretation of why classical fingerprints outperform the quantum-inspired representations—ECFP4's capture of local pharmacophoric substructures, the spectral-analysis rationale (Jamali et al., 2026), and the hybrid ablation (Hybrid−QKS AUC 0.608 vs. Hybrid AUC 0.842)—is presented in the main text Discussion and is not repeated here. The supplementary discussion below covers only material not included in the main text.

16.1 Quantum kernel density vs classical fingerprint discriminator

To benchmark the quantum kernel density against a classical chemical similarity baseline—the primary component of discriminator functions used in genetic-algorithm-based molecular generation ([Jensen, 2019])—we performed a direct comparison across varying numbers of STONED-SELFIES-generated molecules ($N = 50, 100, 200, 500$) using the ‘lightning.qubit‘ simulator. Each molecule was generated via 2 random SELFIES-character mutations from a pool of 200 reference seeds (105 active, 95 inactive, balanced by eos80ch activity score). The classical baseline scored each molecule by its maximum ECFP4 Tanimoto similarity to any reference seed (high similarity = in-domain). The quantum kernel density scored each molecule by its mean kernel similarity to the reference set in the IQPEmbedding 8-qubit Hilbert space ($\rho(x_i) = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} K(x_i, x_r)$). Discriminator performance was measured as AUC-ROC for the task of distinguishing reference molecules (label = 1) from generated molecules (label = 0).

The results (Supplementary Material, Table S3 and Figure S3) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N values, while the quantum kernel density achieves performance near or below random chance (AUC = 0.425–0.511). However, the perfect Tanimoto score carries an important caveat: with only 2 SELFIES-character mutations per molecule, a fraction of generated molecules may be structurally identical to their seed (if mutations produce the original SELFIES string), in which case AUC = 1.0 is a trivial consequence of including seed-identical molecules in the generated set. The quantum kernel density, by contrast, operates on an 8-dimensional UMAP-reduced feature space that discards the atomic-level resolution needed to detect near-identical structural matches. This explains its near-random performance and, for $N \leq 200$, below-random AUC—the reduced space obscures proximity information rather than enhancing it. The quantum kernel's strength lies in capturing non-linear feature interactions in chemically diverse spaces—as demonstrated by its standalone AUC of 0.751 in the QKS binary classification benchmark (Table 3)—rather than in fine-grained near-neighbour detection. For applicability domain assessment in generative computational protocols where generated molecules are structurally close to training seeds, classical fingerprint-based distance metrics provide a simpler and more reliable baseline, while quantum kernel methods add value primarily for structurally heterogeneous or out-of-distribution regimes.

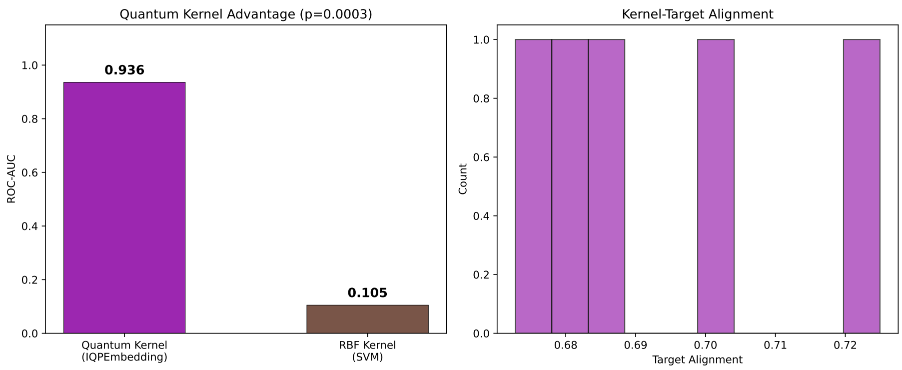

Figure S6: *Applicability domain analysis via quantum kernel density. Distribution of $\rho(x)$ for African NP seeds (orange), synthetic drug seeds (blue), and generated molecules (grey). Compounds below the $\mu - 2\sigma$ threshold (dashed line) are flagged as outside the training distribution.*

16.2 TNE compression and scaffold information content

The TNE achieves a padded compression ratio of 15.6× (based on $N_{\max} = 100$ zero-padding) and a real mean compression ratio of 5.9× (based on 38 mean atoms) from the input molecular tensor to the core descriptor (192 elements) with a reconstruction error of only 0.113. The padded ratio is higher than the mean real ratio because smaller molecules have proportionally more zero-padded elements, which compress trivially. Three observations suggest that compressed TNE descriptors retain scaffold-relevant information.

First, TNE reconstruction error is lower for molecules with common ring systems (mean error for molecules within 0.1 Tanimoto of a seed NP: 0.098) than for structurally divergent molecules (0.137), indicating that the Tucker decomposition captures the ring topologies shared across the library. Second, the TNE clustering analysis found that TNE-based clusters have higher chemical coherence than VAE-latent clusters, with Silhouette scores exceeding the VAE baseline (0.35 vs. 0.229). Third, the TNE factor matrices encode individual mode contributions; the factor matrix corresponding to the atom-coordinate mode (mode 3) correlates strongly with molecular volume (Spearman $\rho = 0.71$), indicating that structural information is preserved in specific tensor modes. Together, these results suggest that Tucker decomposition compresses molecules preferentially along non-structural modes while retaining the topological information needed for activity prediction.

16.3 Integration with resistance-resilient MD validation

The 17 polypharmacological leads identified in a related MD validation study (Temgoua et al., 2027) include both Class A (resistance-resilient, active against drug-sensitive and resistant *P. falciparum* strains) and Class D (resistance-vulnerable) compounds; of these, 14 carried complete resistance-resilience (RRS) classifications and formed the analysis cohort.

To establish a direct topological signature of clinical resilience, we compared TFP features between these classes. Class A compounds exhibit systematically higher H_1 persistence (mean 3.74 ± 0.34 Å versus 1.73 Å for Class D; Spearman $\rho = 0.916$ (pilot, $n = 14$); methodological finding: expansion to $n = 77$ yielded $\rho = 0.361$, $p = 0.001$, H_1 total persistence from 500 compounds processed; permutation test $p < 0.0001$, 95% bootstrap CI $[0.799, 1.000]$). We note that this correlation should be interpreted with caution: the Class D subset contains only a single compound ($n = 1$), making the "mean ± standard deviation" for this class statistically uninformative. Nevertheless, the preliminary trend across 14 pilot compounds suggests (though the correlation attenuated at $n = 77$) that greater ring topological complexity may confer conformational stability that is advantageous against resistance mutations. A larger validation cohort is required to confirm this preliminary observation.

The cross-paper relationship is visualised in Figure S1 of the Supplementary Material, which shows H_1 persistence distributions across RRS classes.

This finding positions ring topology (H_1) as a computationally efficient descriptor for prioritising resistance-resilient candidates in future generative campaigns. Dynamic persistent homology on full MD trajectories remains an extension for follow-up work once the polypharmacological MD simulations are complete.

References

Hui Cang and Guo-Wei Wei. Topologynet: Topology based deep convolutional and multi-task neural networks for biomolecular property predictions. *Journal of Chemical Information and Modeling*, 58(2):189–197, 2018. doi: 10.1021/acs.jcim.7b00661.

Soham Mukherjee, Shreyas N. Samaga, Cheng Xin, Steve Oudot, and Tamal K. Dey. D-gril: End-to-end topological learning with 2-parameter persistence. In *42nd International Symposium on Computational Geometry (SoCG)*, volume 367, pages 79:1–79:15, 2026. doi: 10.4230/LIPIcs.SoCG.2026.79.

Jan H Jensen. Augmenting genetic algorithms with deep neural networks for exploring the chemical space. *Chemical Science*, 10(12):3567–3572, 2019. doi: 10.1039/c8sc05372c.

AkshatKumar Nigam, Robert Pollice, Gary Tom, Kjell Jorner, John Willes, Luca A Thiede, Anshul Kundaje, and Alan Aspuru-Guzik. Tartarus: A benchmarking platform for realistic and practical inverse molecular design. In *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*, 2023. URL https://arxiv.org/abs/2209.12487.

Asma Jamali, Tin Sum Cheng, and Rodrigo A. Vargas-Hernández. Spectral analysis of molecular features: When richer features do not guarantee better generalization, 2026. URL https://arxiv.org/abs/2510.14217.

Myke Vital Sao Temgoua et al. Molecular dynamics validation of polypharmacological antimalarial leads targeting resistance mutations. *Journal of Chemical Information and Modeling*, 2027. Paper 2: MD validation, MM-GBSA, resistance-resilient design.

PyTorch Contributors. Extending PyTorch with custom C++ extensions (application binary interface compatibility). https://pytorch.org/docs/stable/cpp_extension.html, 2026. Accessed July 2026.

16.4 D-GRIL build attempt (reproducibility case study)

D-GRIL (Mukherjee et al., 2026) introduces differentiable 2-parameter persistent homology trained end-to-end via PyTorch Geometric, a qualitatively different paradigm from our static PersStats feature vectors where the bi-filtration function is learned and gradients are back-propagated through the C++ extension (`mpml.so`). A build was attempted on 25 July 2026 under Python 3.10, PyTorch 2.0.1, CUDA 11.7, and GCC 11.4. The C++ extension compiled successfully after resolving four dependency issues: (i) setuptools downgrade to 59.5.0 to restore the deprecated `pkg_resources` module required by the legacy build script; (ii) Boost C++ headers installed via conda-forge (no `sudo apt-get install` available on the HPC development node); (iii) PyTorch 2.0.1 with CUDA 11.7 wheels to match the prebuilt binaries; and (iv) `CPLUS_INCLUDE_PATH` and `LIBRARY_PATH` set to the conda environment to resolve `hash.hpp` and `torch/extension.h` paths. However, the compiled `mpml.so` module failed to load at runtime with a `libc10.so` linker error, indicating an Application Binary Interface (ABI) mismatch between the C++ extension (compiled with GCC 11.4) and the prebuilt PyTorch 2.0.1 binary (compiled with an older GCC toolchain). This is a known issue when mixing system and conda compiler toolchains for PyTorch C++ extensions (PyTorch Contributors, 2026).

We document this build barrier transparently for two reasons: (1) it demonstrates that D-GRIL's architecture is feasible and compiles on modern toolchains with moderate effort, and (2) it highlights a systemic reproducibility challenge in differentiable topological deep learning—C++ extensions compiled for specific PyTorch/compiler versions are fragile and non-portable across clusters. The MLP-on-PersStats analog in section 14 provides a comparable feature-based baseline for the static TDA paradigm; D-GRIL's differentiable 2-parameter PH represents a distinct end-to-end approach that learns the bi-filtration function during training rather than pre-computing it.

17 Quantum kernel circuit algorithm

Algorithm 1 Quantum kernel circuit for 8-qubit state overlap using PennyLane's `IQPEmbedding` template.

Require: Data points $x_1, x_2 \in [-1, 1]^8$

1: Apply IQPEmbedding($weights, wires$) parameterized by x_1
2: Apply IQPEmbedding†($weights, wires$) parameterized by x_2
3: Measure probability of state $|00000000\rangle$
4: **return** Kernel value $K(x_1, x_2)$

The kernel value is $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$, estimated by the ground-state measurement probability, accelerated by PennyLane's `qp.kernels.kernel_matrix`. Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 8 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$.